

	FUNCTIONAL REQUIREMENTS DOCUMENT Spacecraft Status Display and Commanding Software 1.0	Doc. No: SCG-GRCD2-SW-2021-01 Version : 1.0
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Functional Requirements Document

SPAcecraft **S**tatus **D**isplay **A**nd **C**ommanding **S**oftware (SPASDACS) 1.0

GEOSAT RF & Payload Checkout Division-2
Spacecraft Checkout Group

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BANGALORE

	FUNCTIONAL REQUIREMENTS DOCUMENT Spacecraft Status Display and Commanding Software 1.0	Doc. No: SCG-GRCD2-SW-2021-01 Version : 1.0
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	SOFTWARE REQUIREMENTS SPECIFICATION OF Spacecraft Status Display and Commanding Software 1.0	Doc. No: SCG-GRCD2-SW-2021-01 Version: 1.0
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AUTHENTICATION

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CHANGE HISTORY

Version Number	Date	Affected Section, Figure, Table	Nature of change [A,M,D] *	Description
1.0	02/08/2021	-	New	

* A – Addition, D – Deletion, M-Modification

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1. Introduction

This document describes the functional requirements of the Spacecraft Status Display and Commanding Software (SPASDACS). The system provides a graphical interface to visualize spacecraft subsystem telemetry using interactive block diagrams.

2. Scope

SPASDACS enables engineers to design subsystem block diagrams and monitor spacecraft telemetry in real time. The software will be used during spacecraft checkout and testing activities.

3. System Overview

The system displays spacecraft subsystems as graphical blocks. Telemetry data is linked to these blocks and the visual attributes change dynamically based on telemetry values. This allows engineers to quickly assess subsystem status.

4. User Roles

- Admin
 - Create and edit subsystem diagrams
 - Associate telemetry parameters with diagram elements
 - Configure visualization rules and alerts
- User / Guest
 - View real-time subsystem diagrams
 - Monitor telemetry alerts and system status

5. Functional Requirements

Requirement ID	Description
FR-01	The system shall allow users to create subsystem block diagrams using basic shapes such as rectangles, circles, ellipses, triangles and lines.
FR-02	The system shall allow users to import custom shapes or images into the diagram.
FR-03	The system shall allow telemetry parameters to be associated with diagram elements.
FR-04	The system shall update visual attributes such as color, visibility, and size based on telemetry values.
FR-05	The system shall display real-time telemetry updates.
FR-06	The system shall support multiple schematic pages.

FR-07	The system shall display telemetry values when the user hovers over a subsystem block.
FR-08	The system shall generate visual and audio alerts when telemetry limits are exceeded.
FR-09	The system shall notify users when telemetry data links are lost.
FR-10	The system shall visually indicate subsystem state changes.

Table 1 Functional Requirements

6. User Workflow

1. User launches the SPASDACS application.
2. User loads or creates a subsystem block diagram.
3. Telemetry parameters are linked to diagram elements.
4. The system subscribes to telemetry data streams.
5. Incoming telemetry updates modify the visual state of blocks.
6. Alerts are generated if telemetry values cross defined limits.

7. Graphical User Interface Overview

The graphical interface consists of the following components:

1. Diagram editor for creating subsystem block diagrams
2. Shape palette for inserting graphical components
3. Telemetry association panel
4. Real-time monitoring display
5. Alert notification panel

8. Future Enhancements

1. Direct telecommand capability from graphical interface
2. Advanced telemetry analytics
3. Custom subsystem component libraries
4. Remote web-based monitoring dashboards